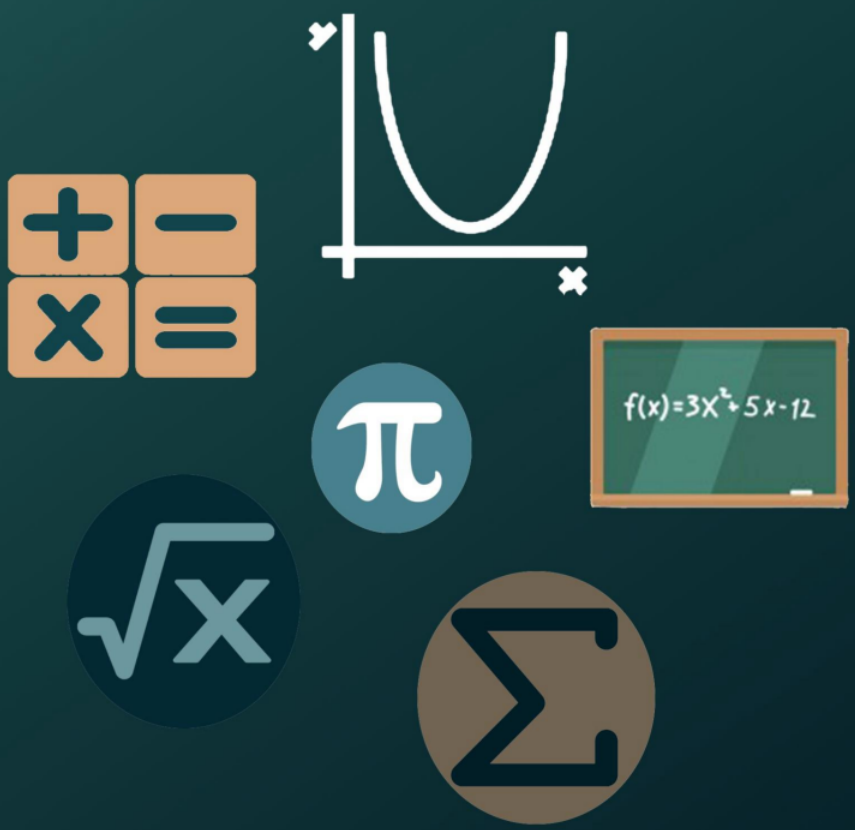
MATHEMATICS

# Inverse Trigonometric Function




SHEET

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***Note***

*Detailed solutions are available on the eSaral App.*

# *Module Description*




For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

*"Practice makes a man perfect"*

**1. Concept builder – 1 & 1A**

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

**2. Brain Booster – 2 & 2A**

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

**3. Simulator – JM & JA**

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

# *Home Work Index*

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

| Sr. No. | Topics | Ex - 1 | Ex - 1A | Ex - 2 | Ex - 2A |
|---|---|---|---|---|---|
| 1 | Domain, Range & Graphs of arc sinx | 1-5 | 19, 20 | 1 - 5 | 1 - 2, 10, 19, 25 - 26 |
| 2 | Domain, Range & Graphs of arc cosx | | | | |
| 3 | Domain, Range & Graphs of arc tanx | | | | |
| 4 | Domain, Range & Graphs of arc cotx | | | | |
| 5 | Domain, Range & Graphs of arc secx & arc cosecx | | | | |
| 6 | Properties of Inverse Trigonometric Functions : 1 | 6, 7 | 1 - 2, 21 - 22 | | |
| 7 | Properties of Inverse Trigonometric Functions : 2 & 3 | 8, 9 | | | 11 |
| 8 | Properties of Inverse Trigonometric Functions : 4 | 10-13 | 4, 5 | 6 | |
| 9 | Properties of Inverse Trigonometric Functions : 5 | 14-20 | 6, 17 - 18 | 7 - 8 | 3, 27 |
| 10 | Properties of Inverse Trigonometric Functions : 6 & 7 | 21-23 | 3, 7 | 9 - 11 | 4 |
| 11 | Properties of Inverse Trigonometric Functions : 8 | 24-26 | 8 - 9, 11 - 13 | 12 - 13 | 12, 20 |
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| 14 | Simultaneous In-equations Involving ITF | 43, 44 | | 21 - 25 | 9, 30 - 31 |
| 15 | Summation of Series using ITF | | | 26 - 29 | 21, 32 |
| 16 | After complete chapter | | 14 - 16 | | 16 - 18, 22 - 24, 33 - 36 |

## Exercise - 1 — Concept Builder

### SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

### Domain and Range of Inverse Trigo Functions & Graphs of arc sinx

**Q1.** If $\sum_{i=1}^{20} \sin^{-1} x_i = 10\pi$ then $\sum_{i=1}^{20} x_i$ is equal to

(A) 20 (B) 10

(C) 0 (D) None of these

**Q2.** The value of $\tan\left(\frac{1}{2}\cos^{-1}\left(\frac{\sqrt{5}}{3}\right)\right)$ is

(A) $\frac{3+\sqrt{5}}{2}$ (B) $3+\sqrt{5}$

(C) $\frac{1}{2}(3-\sqrt{5})$ (D) None of these

**Q3.** Let $f(x) = \cot^{-1} x + \operatorname{cosec}^{-1} x$. Then f(x) is real for

(A) $x \in [-1,1]$

(B) $x \in (-\infty,-1] \cup [1,\infty)$

(C) $x \in (-\infty,\infty)$

(D) None of these

**Q4.** $\sec^{-1}(\sin x)$ is real , if

(A) $x \in (-\infty,\infty)$

(B) $x \in [-1,1]$

(C) $x = (2n+1)\frac{\pi}{2}, n \in I$

(D) $x = n\pi, n \in Z$

**Q5.** Range of $f(x) = \sin^{-1} x + \tan^{-1} x + \sec^{-1} x$, is

(A) $\left(\frac{\pi}{4},\frac{3\pi}{4}\right)$ (B) $\left[\frac{\pi}{4},\frac{3\pi}{4}\right]$

(C) $\left\{\frac{\pi}{4},\frac{3\pi}{4}\right\}$ (D) None of these

### Properties of Inverse Trigonometric Functions : 1

**Q6.** The value of $\cos\left[\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)+\frac{\pi}{6}\right]$ is

(A) 1 (B) –1

(C) 0 (D) None of these

**Q7.** The value of $\sin^2\left(\cos^{-1}\frac{1}{2}\right)+\cos^2\left(\sin^{-1}\frac{1}{3}\right)$ is -

(A) $\frac{17}{36}$ (B) $\frac{59}{36}$

(C) $\frac{36}{59}$ (D) None of these

### Properties of Inverse Trigonometric Functions : 2 & 3

**Q8.** If $\sin^{-1}\left(\frac{x}{5}\right) + \operatorname{cosec}^{-1}\left(\frac{5}{4}\right) = \frac{\pi}{2}$ then a value of x is-

(A) 1 (B) 3

(C) 4 (D) 5

**Q9.** The value of

$\sin\left[\tan^{-1}(-\sqrt{3}) + \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right]$ is

(A) 1 (B) –1

(C) 0 (D) None of these

## Properties of Inverse Trigonometric Functions : 4

**Q10.** Which one of the following is true?

(A) $\sin(\cos^{-1} x) = \cos(\sin^{-1} x)$

(B) $\sec(\tan^{-1} x) = \tan(\sec^{-1} x)$

(C) $\cos(\tan^{-1} x) = \tan(\cos^{-1} x)$

(D) $\tan(\sin^{-1} x) = \sin(\tan^{-1} x)$

**Q11.** The value of

$$\sin\left(\sin^{-1}\frac{1}{3}+\sec^{-1}3\right)+\cos\left(\tan^{-1}\frac{1}{2}+\tan^{-1}2\right)$$

is :

(A) 1 (B) 2 (C) 3 (D) 4

**Q12.** If $\sin^{-1} x = \frac{\pi}{5}$, for some $x \in (-1, 1)$, then the value of $\cos^{-1} x$ is

(A) $\frac{3\pi}{10}$ (B) $\frac{5\pi}{10}$ (C) $\frac{7\pi}{10}$ (D) $\frac{9\pi}{10}$

**Q13.** The value of $\tan\left\{\cos^{-1}\left(-\frac{2}{7}\right)-\frac{\pi}{2}\right\}$ is

(A) $\frac{2}{3\sqrt{5}}$ (B) $\frac{2}{3}$

(C) $\frac{1}{\sqrt{5}}$ (D) $\frac{4}{\sqrt{5}}$

## Properties of Inverse Trigonometric Functions : 5

**Q14.** The value of $\cot^{-1} 9 + \operatorname{cosec}^{-1}\frac{\sqrt{41}}{4}$ is given by :

(A) 0 (B) $\frac{\pi}{4}$

(C) $\tan^{-1} 2$ (D) $\frac{\pi}{2}$

**Q15.** The value of $\cot\left(\operatorname{cosec}^{-1}\frac{5}{3}+\tan^{-1}\frac{2}{3}\right)$ is equal to-

(A) $\frac{6}{17}$ (B) $\frac{3}{17}$

(C) $\frac{4}{17}$ (D) $\frac{5}{17}$

**Q16.** $\tan^{-1}\left(\frac{x}{y}\right)-\tan^{-1}\frac{x-y}{x+y}$ is

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$

(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$ or $-\frac{3\pi}{4}$

**Q17.** If $\alpha, \beta$ are the roots of the equation $6x^2 - 5x + 1 = 0$, then the value of $\tan^{-1}\alpha + \tan^{-1}\beta$ is :

(A) 0 (B) $\pi/4$

(C) 1 (D) $\pi/2$

**Q18.** If $\tan^{-1} a + \tan^{-1} b = \sin^{-1} 1 - \tan^{-1} c$, then :

(A) $a + b + c = abc$

(B) $ab + bc + ca = abc$

(C) $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}-\frac{1}{abc}=0$

(D) $ab + bc + ca = a + b + c$

**Q19.** If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$, then the value of $x + y + z$ is :

(A) $-xyz$ (B) $xyz$ (C) $\frac{1}{xyz}$ (D) 0

**Q20.** If the non-zero numbers $x$, $y$, $z$ are in arithmetic progression and $\tan^{-1} x$, $\tan^{-1} y$, $\tan^{-1} z$ are also in arithmetic progression, then -

(A) $x = y = z$ (B) $xy = yz$

(C) $x^2 = yz$ (D) $z^2 = xy$

## Properties of Inverse Trigonometric Functions : 6 & 7

**Q21.** $\sin^{-1}\frac{1}{\sqrt{5}}+\cot^{-1}3$ is equal to :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$

(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

**Q22.** If $x \in \left(\frac{3\pi}{2}, 2\pi\right)$; then the value of the expression $\sin^{-1}\left(\cos\left(\cos^{-1}(\cos x)+\sin^{-1}(\sin x)\right)\right)$ is :

(A) $-\frac{\pi}{2}$ (B) $\frac{\pi}{2}$

(C) 0 (D) $\pi$

**Q23.** If $\cos^{-1}x-\cos^{-1}\frac{y}{2}=\alpha$, then $4x^2-4xy\cos\alpha+y^2$ is equal to-

(A) $2\sin 2\alpha$ (B) 4

(C) $4\sin^2\alpha$ (D) $-4\sin^2\alpha$

## Properties of Inverse Trigonometric Functions : 8

**Q24.** The value of $\sin^{-1}(\sin 10)$ is :

(A) $3\pi-10$ (B) $10-3\pi$

(C) 10 (D) None of these

**Q25.** The value of $\sin\left[\cot^{-1}\left(\cot\frac{17\pi}{3}\right)\right]$ is :

(A) $-\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{3}}{2}$

(C) $\frac{1}{\sqrt{2}}$ (D) None of these

**Q26.** The value of $\cos^{-1}\left(\cos\frac{5\pi}{3}\right)+\sin^{-1}\left(\sin\frac{5\pi}{3}\right)$ is

(A) $\frac{\pi}{2}$ (B) $\frac{5\pi}{3}$

(C) $\frac{10\pi}{3}$ (D) 0

## Simplification & Transformation of Inverse Functions by Elementary Substitution & their Graphs

**Q27.** $\tan(\cos^{-1}x)$ is equal to

(A) $\pm\sqrt{\frac{1-x^2}{x}}, x\neq 0$ (B) $\frac{\sqrt{1+x^2}}{x}, x\neq 0$

(C) $\frac{x}{\sqrt{1+x^2}}$ (D) $\frac{\sqrt{1-x^2}}{x}, x\neq 0$

**Q28.** For $0\le\cos^{-1}x\le\pi$ and $-\frac{\pi}{2}\le\sin^{-1}x\le\frac{\pi}{2}$, the value of $\cos(\sin^{-1}x+2\cos^{-1}x)$ at $x=\frac{1}{5}$ is-

(A) $\frac{-2\sqrt{6}}{5}$ (B) $-\frac{\sqrt{6}}{5}$

(C) $\frac{2\sqrt{6}}{5}$ (D) $\frac{\sqrt{6}}{5}$

**Q29.** $3\tan^{-1}a$ is equal to :

(A) $\tan^{-1}\frac{3a+a^3}{1+3a^2}$ (B) $\tan^{-1}\frac{3a-a^3}{1+3a^2}$

(C) $\tan^{-1}\frac{3a+a^3}{1-3a^2}$ (D) $\tan^{-1}\frac{3a-a^3}{1-3a^2}$

**Q30.** The value of $\sin(\cot^{-1}x)$ is :

(A) $(1+x^2)^{-1/2}$ (B) $(1+x^2)^{1/2}$

(C) $(1+x^2)^{3/2}$ (D) $(1+x^2)^{-3/2}$

**Q31.** $\tan\left(\frac{\pi}{4}+\frac{1}{2}\cos^{-1}\frac{a}{b}\right)+\tan\left(\frac{\pi}{4}-\frac{1}{2}\cos^{-1}\frac{a}{b}\right)=$

(A) $\frac{2a}{b}$ (B) $\frac{a}{b}$

(C) $\frac{b}{a}$ (D) $\frac{2b}{a}$

**Q32.** Let $f:(-1,1)\to B$, be a function defined by $f(x)=\tan^{-1}\frac{2x}{1-x^2}$, then $f$ is both one-one and onto when B is the interval.

(A) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ (B) $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

(C) $\left[0,\frac{\pi}{2}\right)$ (D) $\left(0,\frac{\pi}{2}\right)$

## Equations Involving Inverse Trigonometric Functions

**Q33.** If $\cos^{-1}x-\sin^{-1}x=0$, then x is equal to-

(A) $\pm\frac{1}{\sqrt{2}}$ (B) 1

(C) $\pm\frac{1}{\sqrt{3}}$ (D) $\frac{1}{\sqrt{2}}$

**Q34.** Solution of the equation

$\cot^{-1}x+\sin^{-1}\frac{1}{\sqrt{5}}=\frac{\pi}{4}$ is

(A) x =3 (B) $x=\frac{1}{\sqrt{5}}$

(C) x = 0 (D) None of these

**Q35.** For which value of x,
$\sin(\cot^{-1}(x+1))=\cos(\tan^{-1}x)$

(A) 1/2 (B) 0

(C) 1 (D) –1/2

**Q36.** If $\cos^{-1}\frac{3}{5}-\sin^{-1}\frac{4}{5}=\cos^{-1}x$, then x =

(A) 0 (B) 1

(C) –1 (D) None of these

**Q37.** If $\tan^{-1}3+\tan^{-1}x=\tan^{-1}8$, then x =

(A) 5 (B) 1/5

(C) 5/14 (D) 14/5

**Q38.** If $\tan^{-1}(x-1)+\tan^{-1}x+\tan^{-1}(x+1)=\tan^{-1}3x$, then $x$ is

(A) $\pm\frac{1}{2}$ (B) $0,\frac{1}{2}$

(C) $0,-\frac{1}{2}$ (D) $0,\pm\frac{1}{2}$

**Q39.** If $\tan^{-1}1/3+\tan^{-1}3/4-\tan^{-1}x/3=0$, then $x$ is equal to :

(A) $\frac{7}{3}$ (B) 3

(C) $\frac{11}{3}$ (D) $\frac{13}{3}$

**Q40.** The number of real solution of

$\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{x^2+x+1}=\frac{\pi}{2}$ is-

(A) Zero (B) One

(C) Two (D) Infinite

**Q41.** The solution of

$\sin^{-1}\left(\frac{2a}{1+a^2}\right)-\cos^{-1}\left(\frac{1-b^2}{1+b^2}\right)$

$=\tan^{-1}\left(\frac{2x}{1-x^2}\right)$ is-

(A) $\frac{a-b}{1-ab}$ (B) $\frac{1+ab}{a-b}$

(C) $\frac{ab-1}{a+b}$ (D) $\frac{a-b}{1+ab}$

**Q42.** If $\sin^{-1}\left(x-\frac{x^2}{2}+\frac{x^3}{4}-\cdots\right)+\cos^{-1}\left(x^2-\frac{x^4}{2}+\frac{x^6}{4}-\cdots\right)=\frac{\pi}{2}$ for $0<|x|<\sqrt{2}$, then x equals-

(A) $\frac{1}{2}$ (B) 1

(C) $-\frac{1}{2}$ (D) –1

**Simultaneous In-equations Involving ITF**

**Q43.** $\tan^{-1}x > \cot^{-1}x$ holds for

(A) $x > 1$ (B) $x < 1$

(C) $x = 1$ (D) all values of x

**Q44.** If $\tan^{-1}\frac{x}{\pi}<\frac{\pi}{3}, x\in N$, then the largest integral value of x is

(A) 2 (B) 5

(C) 7 (D) None of these

## *Exercise - 1A* *Concept Builder*

### ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** $\sin(\cot^{-1}x) =$

(A) $\sqrt{(1+x^2)}$ (B) $\frac{1}{\sqrt{\left(1+|x|^2\right)}}$

(C) $(1+x^2)^{-3/2}$ (D) $(1+x^2)^{-1/2}$

**Q2.** $\tan(\cos^{-1}x)$ is equal to-

(A) $\frac{\sqrt{1-x^2}}{x}$ (B) $\frac{x}{1+x^2}$

(C) $\frac{\sqrt{1+x^2}}{x}$ (D) $\sqrt{1-x^2}$

**Q3.** If $\cos^{-1}\left(\frac{x}{a}\right)+\cos^{-1}\left(\frac{y}{b}\right)=\alpha$ then $\frac{x^2}{a^2}-\frac{2xy}{ab}\cos\alpha+\frac{y^2}{b^2}$ is equal to :

(A) $\cos^2\alpha$ (B) $\sin^2\alpha$

(C) $\tan^2\alpha$ (D) $\cot^2\alpha$

**Q4.** The value of $\cos(\sec^{-1}x + \operatorname{cosec}^{-1}x)$, $|x|\geq 1$ is-

(A) 1 (B) –1

(C) 0 (D) None of these

**Q5.** If $a \leq \tan^{-1}x+\cot^{-1}x+\sin^{-1}x \leq b$, then $(x \in R)$

(A) $a = 0$ (B) $b = 2\pi$

(C) $a = -2$ (D) $b = \pi$

**Q6.** $\tan^{-1}\left(\frac{1}{4}\right)+\tan^{-1}\left(\frac{2}{9}\right)=$

(A) $\frac{1}{2}\tan^{-1}\left(\frac{3}{5}\right)$ (B) $\sin^{-1}\frac{1}{\sqrt{5}}$

(C) $\tan^{-1}\left(\frac{3}{5}\right)$ (D) $\tan^{-1}\left(\frac{1}{2}\right)$

**Q7.** If $\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{5}{13}=\sin^{-1}x$, then $x =$

(A) $\frac{41}{56}$ (B) $\frac{56}{65}$ (C) $\frac{13}{41}$ (D) $\frac{56}{41}$

**Q8.** The principal value of $\sin^{-1}\left[\sin\left(\frac{2\pi}{3}\right)\right]$ is-

(A) $-\frac{\pi}{6}$ (B) $\frac{2\pi}{3}$ (C) $-\frac{\pi}{3}$ (D) $\frac{\pi}{3}$

**Q9.** The principal value of $\tan^{-1}\tan\left(\frac{3\pi}{4}\right)$ is-

(A) $-\frac{\pi}{6}$ (B) $\frac{\pi}{3}$

(C) $-\frac{\pi}{4}$ (D) $\frac{3\pi}{4}$

**Q10.** Solution of the equation

$\tan^{-1}2x+\tan^{-1}3x=n\pi+\frac{3\pi}{4}$ is

(A) $x=1$ (B) $x=\frac{1}{6}$

(C) $x=-\frac{1}{6}$ (D) $x=-1$

## PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

**Paragraph # 1 (Ques.11 to 13) :**

Principal values for inverse circular functions

| $x < 0$ | $x \geq 0$ |
|---|---|
| $-\frac{\pi}{2} \leq \sin^{-1}x < 0$ | $0 \leq \sin^{-1}x \leq \frac{\pi}{2}$ |
| $\frac{\pi}{2} < \cos^{-1}x \leq \pi$ | $0 \leq \cos^{-1}x \leq \frac{\pi}{2}$ |
| $-\frac{\pi}{2} < \tan^{-1}x < 0$ | $0 \leq \tan^{-1}x < \frac{\pi}{2}$ |
| $\frac{\pi}{2} < \cot^{-1}x < \pi$ | $0 < \cot^{-1}x \leq \frac{\pi}{2}$ |
| $\frac{\pi}{2} < \sec^{-1}x \leq \pi$ | $0 \leq \sec^{-1}x < \frac{\pi}{2}$ |
| $-\frac{\pi}{2} \leq \operatorname{cosec}^{-1}x < 0$ | $0 < \operatorname{cosec}^{-1}x \leq \frac{\pi}{2}$ |

**Q11.** The principal value of

$\sin^{-1}\left(\sin\frac{4\pi}{3}\right)+\cos^{-1}\left(\cos\frac{4\pi}{3}\right)$ is

(A) $\frac{8\pi}{3}$ (B) $\frac{4\pi}{3}$

(C) $\frac{2\pi}{3}$ (D) $\frac{\pi}{3}$

**Q12.** The principal value of
$\sin^{-1}(\sin 5) - \cos^{-1}(\cos 5)$ is
(A) 0 (B) $2\pi - 10$
(C) $10 - 4\pi$ (D) $3\pi - 10$

**Q13.** The value of
$\sin^{-1}[\cos\{\cos^{-1}(\cos x) + \sin^{-1}(\sin x)\}]$,
where $x \in \left(\frac{\pi}{2}, \pi\right)$ is-

(A) $\frac{\pi}{2}$ (B) $-\pi$

(C) $\pi$ (D) $-\frac{\pi}{2}$

**Paragraph # 2 (Ques.14 to 16)**

Consider $f(x) = \sin^{-1}x + \tan^{-1}x + \sqrt{x+1}$ & let g(x) be its inverse.

**Q14.** Number of integral values of k such that $f(x) = k$ has atleast one real solution is equal to -
(A) 3 (B) 4
(C) 5 (D) 6

**Q15.** If $g(0) = \ell$, then -

(A) $-1 < \ell < -\frac{1}{2}$ (B) $-\frac{1}{2} < \ell < 0$

(C) $0 < \ell < \frac{1}{2}$ (D) $-1 < \ell < -\frac{3}{4}$

**Q16.** Number of solutions of the equation $f(|x|) = 2$ is equal to -
(A) 0 (B) 1 (C) 2 (D) 4

## MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

**Q17.** Observe the following columns :

| **Column-I** | **Column-II** |
|---|---|
| (A) $\tan^{-1}3 + \tan^{-1}4$ | (P) $\frac{\pi}{2}$ |
| (B) $\tan^{-1}(1/3) + \tan^{-1}(1/4)$ | (Q) $\pi - \tan^{-1}(7/11)$ |
| (C) $\sin^{-1}(1/3) + \cos^{-1}(1/3)$ | (R) $\tan^{-1}(7/11)$ |

**Q18.** $x \geq 0$, $y \geq 0$, $z \geq 0$ and $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = k$, the possible value(s) of k, if

| **Column-I** | **Column-II** |
|---|---|
| (A) $xy + yz + zx = 1$, then | (P) $k = \frac{\pi}{2}$ |
| (B) $x + y + z = xyz$, then | (Q) $k = \pi$ |
| (C) $x^2 + y^2 + z^2 = 1$ and $x + y + z = \sqrt{3}$, then | (R) $k = 0$ |
| (D) $x = y = z$ and $xyz \geq 3\sqrt{3}$, then | (S) $k = \frac{7\pi}{6}$ |

## SUBJECTIVE TYPE

**Q19.** Find the principal value of $\cos^{-1}\frac{\sqrt{3}}{2}$ and $\cos^{-1}\left(-\frac{1}{2}\right)$.

**Q20.** Find the principal value of $\cot^{-1}\left(-\frac{1}{\sqrt{3}}\right)$

**Q21.** Prove that :
$\sec^2(\tan^{-1}2) + \operatorname{cosec}^2(\cot^{-1}3) = 15$

**Q22.** Evaluate the following :

(i) $\sin(\cos^{-1}3/5)$

(ii) $\cos(\tan^{-1}3/4)$

(iii) $\sin\left(\frac{\pi}{2} - \sin^{-1}\left(-\frac{1}{2}\right)\right)$

(iv) $\tan\left\{2\tan^{-1}\frac{1}{5} - \frac{\pi}{4}\right\}$

## Exercise - 2 — Brain Booster

### SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

**Q1.** The domain of the function $f(x)=\sqrt{e^{\sin^{-1}(\log_{16}x^2)}}$ is -

(A) $\left[\frac{1}{4},4\right]$ (B) $\left[-4,-\frac{1}{4}\right]\cup\left[\frac{1}{4},4\right]$

(C) $\left[-4,-\frac{1}{4}\right]$ (D) None of these

**Q2.** Let $f:R\to\left[0,\frac{\pi}{2}\right)$ defined by $f(x)=tan^{-1}(x^2+x+a)$, then the set of values of a for which 'f' is onto is -

(A) $[0,\infty)$ (B) $[2,1]$

(C) $\left[\frac{1}{4},\infty\right)$ (D) $\left\{\frac{1}{4}\right\}$

**Q3.** If $\left[\sin^{-1}\cos^{-1}\sin^{-1}\tan^{-1}x\right]=1$, where [x] denotes the greatest integer function, then x lies in.

(A) [tan sin cos 1, tan sin cos sin 1]

(B) (tan sin cos 1, tan sin cos sin 1)

(C) [–1, 1]

(D) [sin cos tan 1, sin cos sin tan 1]

**Q4.** $\cos\left[\tan^{-1}\left[\sin(\cot^{-1}x)\right]\right]=$

(A) $\sqrt{\frac{x^2+2}{x^2+3}}$ (B) $\sqrt{\frac{x^2+2}{x^2+1}}$

(C) $\sqrt{\frac{x^2+1}{x^2+2}}$ (D) $\frac{1}{\sqrt{1+x^2}}$

**Q5.** $\sin\left[\cot^{-1}\left\{\cos\left(\tan^{-1}x\right)\right\}\right]=$

(A) $\sqrt{\frac{1+x^2}{2+x^2}}$ (B) $\sqrt{\frac{1-x^2}{2+x^2}}$

(C) $\sqrt{\frac{1+x^2}{2-x^2}}$ (D) $\sqrt{\frac{2+x^2}{1+x^2}}$

**Q6.** The least and greatest values of $(\sin^{-1}x)^3+(\cos^{-1}x)^3$ are respectively.

(A) $-\frac{\pi}{2},\frac{\pi}{2}$ (B) $-\frac{\pi^3}{8},\frac{\pi^3}{8}$

(C) $\frac{\pi^3}{32},\frac{7\pi^3}{8}$ (D) $-\frac{\pi^3}{32},\frac{\pi^3}{32}$

**Q7.** In a $\Delta ABC$, if $A=\tan^{-1}2$ and $B=\tan^{-1}3$, then C =

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{\pi}{2}$

**Q8.** The value of $\tan^{-1}\sqrt{\frac{a\lambda}{bc}}+\tan^{-1}\sqrt{\frac{b\lambda}{ca}}+\tan^{-1}\sqrt{\frac{c\lambda}{ab}}$ where a, b, c $\in R^+$ and $\lambda\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right)=1$, is equal to :-

(A) $\frac{\pi}{4}$ (B) $\pi$

(C) $\frac{\pi}{2}$ (D) None of these

**Q9.** The value of $\cos^{-1}\sqrt{\frac{2}{3}}-\cos^{-1}\frac{\sqrt{6}+1}{2\sqrt{3}}$ is equal to -

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{6}$

**Q10.** If $\cos^{-1}x+\cos^{-1}y+\cos^{-1}z=\pi$,

then $x^2+y^2+z^2+2xyz=$

(A) 0 (B) 1

(C) –1 (D) None of these

**Q11.** If $\cos^{-1}\left(\frac{p}{a}\right)+\cos^{-1}\left(\frac{q}{b}\right)=\alpha$, then

$\frac{p^2}{a^2}+k\cos\alpha+\frac{q^2}{b^2}=\sin^2\alpha$, where k =

(A) $\frac{2pq}{ab}$ (B) $-\frac{2pq}{ab}$

(C) $\frac{pq}{ab}$ (D) $-\frac{pq}{ab}$

**Q12.** The value of

$\cot^{-1}\left[\frac{\sqrt{1-\sin x}+\sqrt{1+\sin x}}{\sqrt{1-\sin x}-\sqrt{1+\sin x}}\right]$ is

$\left(x\in\left(0,\frac{\pi}{2}\right)\right)$

(A) $\pi-x$ (B) $2\pi-x$

(C) $\frac{x}{2}$ (D) $\pi-\frac{x}{2}$

**Q13.** The value of

$\tan\left(\sin^{-1}\left(\cos\left(\sin^{-1}x\right)\right)\right)\tan\left(\cos^{-1}\left(\sin\left(\cos^{-1}x\right)\right)\right)$,

$x\in(0,1)$

is equal to

(A) 0 (B) 1

(C) –1 (D) None of these

**Q14.** The value of $\tan\left[2\tan^{-1}\left(\frac{1}{5}\right)-\frac{\pi}{4}\right]$ is

(A) $\frac{7}{17}$ (B) $-\frac{7}{17}$

(C) $\frac{2}{3}$ (D) $-\frac{2}{3}$

**Q15.** $\tan^{-1}\left[\cos\left(2\tan^{-1}\frac{3}{4}\right)+\sin\left(2\cot^{-1}\frac{1}{2}\right)\right]$ is

(A) not real

(B) equal to $\pi/4$

(C) greater then $\pi/4$

(D) less than $\pi/4$

**Q16.** Let $f:\left[\frac{1}{2},1\right]\to[-1,1]$ is given by

$f(x)=4x^3-3x$, then $f^{-1}(x)$ is given by :

(A) $\cos\left(\frac{1}{3}\cos^{-1}x\right)$

(B) $3\cos\left(\sin^{-1}x\right)$

(C) $3\sin^{-1}\left(\cos x\right)$

(D) None of these

**Q17.** If $\cot^{-1}\sqrt{\cos\alpha}-\tan^{-1}\sqrt{\cos\alpha}=u$,

then sin u =

(A) $\tan^2\alpha$ (B) $\tan 2\alpha$

(C) $\tan^2\left(\frac{\alpha}{2}\right)$ (D) $\cot^2\left(\frac{\alpha}{2}\right)$

**Q18.** Solution of the equation

$\tan(\cos^{-1}x)=\sin\left(\cot^{-1}\frac{1}{2}\right)$ is -

(A) $x=\pm\frac{\sqrt{7}}{3}$ (B) $x=\pm\frac{\sqrt{5}}{3}$

(C) $x=\pm\frac{3\sqrt{5}}{2}$ (D) None of these

**Q19.** If $\tan^{-1}\frac{1}{1+2x}+\tan^{-1}\frac{1}{4x+1}=\tan^{-1}\frac{2}{x^2}$,

then x =

(A) 0 (B) 3

(C) $-\frac{2}{3}$ (D) all of these

**Q20.** The set of values of a for which

$x^2+ax+\sin^{-1}\left(x^2-4x+5\right)$

$+\cos^{-1}\left(x^2-4x+5\right)=0$

has at least one solution is -

(A) $(-\infty,-\sqrt{2\pi}]\cup[\sqrt{2\pi},\infty)$

(B) $\left(-\infty,-\sqrt{2\pi}\right)\cup\left(\sqrt{2\pi},\infty\right)$

(C) R

(D) None of these

**Q21.** If $\left|\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right|<\frac{\pi}{3}$, then

(A) $x\in\left[-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right]$

(B) $x\in\left(-\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$

(C) $x\in\left(0,\frac{1}{\sqrt{3}}\right)$

(D) None of these

**Q22.** The complete solution set of

$\sin^{-1}(\sin 5)>x^2-4x$, is

(A) $|x-2|<\sqrt{9-2\pi}$

(B) $|x-2|>\sqrt{9-2\pi}$

(C) $|x|<\sqrt{9-2\pi}$

(D) $|x|>\sqrt{9-2\pi}$

**Q23.** Complete solution set of $\tan^2\left(\sin^{-1}x\right)>1$ is -

(A) $\left(-1,-\frac{1}{\sqrt{2}}\right)\cup\left(\frac{1}{\sqrt{2}},1\right)$

(B) $\left(-\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)-\{0\}$

(C) $(-1,1)-\{0\}$

(D) None of these

**Q24.** If $\left[\sin^{-1}x\right]+\left[\cos^{-1}x\right]=0$, where 'x' is a non-negative real number and [.] denotes the greatest integer function, then complete set of values of x is -

(A) $(\cos 1,1)$ (B) $(-1,\cos 1)$

(C) $(\sin 1,1)$ (D) $(\cos 1,\sin 1)$

**Q25.** Solution set of the equation

$\left[\cot^{-1}x\right]+2\left[\tan^{-1}x\right]=0$,

where [.] denotes the greatest integer function, is equal to -

(A) $(0,\cot 1)$ (B) $(0,\tan 1)$

(C) $(\tan 1,\infty)$ (D) $(\cot 1,\tan 1)$

**Q26.** $\tan^{-1}\left(\frac{a_1x-y}{a_1y+x}\right)+\tan^{-1}\left(\frac{a_2-a_1}{a_1a_2+1}\right)+\tan^{-1}$

$\left(\frac{a_3-a_2}{a_2a_3+1}\right)+....+\tan^{-1}\left(\frac{a_n-a_{n-1}}{a_{n-1}a_n+1}\right)$

$+\tan^{-1}\frac{1}{a_n}=\{a_1,a_2,.....,a_n,x,y>0\}$

(A) $\tan^{-1}xy$ (B) $\tan^{-1}\frac{x}{y}$

(C) $\tan^{-1}\frac{y}{x}$ (D) None of these

**Q27.** If $a_1,a_2,a_3,...,a_n$ is an AP with common difference d, then ($a_i>0$, i = 1, 2, 3, ..... n)

$\tan\left[\tan^{-1}\left(\frac{d}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{d}{1+a_2a_3}\right)+\right.$

$\left....+\tan^{-1}\left(\frac{d}{1+a_{n-1}a_n}\right)\right]$

is equal to -

(A) $\frac{(n-1)d}{a_1+a_n}$ (B) $\frac{(n-1)d}{1+a_1a_n}$

(C) $\frac{nd}{1+a_1a_n}$ (D) $\frac{a_n-a_1}{a_n+a_1}$

**Q28.** $\sum_{r=1}^{n}\tan^{-1}\left(\frac{2^{r-1}}{1+2^{2r-1}}\right)$ is equal to

(A) $\tan^{-1}\left(2^{n}\right)$

(B) $\tan^{-1}\left(2^{n}\right)-\frac{\pi}{4}$

(C) $\tan^{-1}\left(2^{n+1}\right)$

(D) $\tan^{-1}\left(2^{n+1}\right)-\frac{\pi}{4}$

**Q29.** $\sum_{r=1}^{n}\sin^{-1}\left(\frac{\sqrt{r}-\sqrt{r-1}}{\sqrt{r(r+1)}}\right)$ is equal to

(A) $\tan^{-1}\left(\sqrt{n}\right)-\frac{\pi}{4}$

(B) $\tan^{-1}\left(\sqrt{n+1}\right)-\frac{\pi}{4}$

(C) $\tan^{-1}\left(\sqrt{n}\right)$

(D) $\tan^{-1}\left(\sqrt{n+1}\right)$

## *Exercise - 2A* — *Brain Booster*

### ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

**Q1.** The range of the function

$y=\sin^{-1}\left(\frac{x^2}{1+x^2}\right)$ is -

(A) $\left(0,\frac{\pi}{2}\right)$ (B) $\left[0,\frac{\pi}{2}\right)$

(C) $\left[0,\frac{\pi}{2}\right]$ (D) None of these

**Q2.** The domain of the function

$f(x)=\cos^{-1}(x+[x])$,

where $[\cdot]$ denotes the greatest integer function, is -

(A) [–1, 1]

(B) [0, 1)

(C) (–1, 0)

(D) None of these

**Q3.** The value of

$$\tan^{-1}\sqrt{\frac{a(a+b+c)}{bc}}+\tan^{-1}\sqrt{\frac{b(a+b+c)}{ca}}+\tan^{-1}\sqrt{\frac{c(a+b+c)}{ab}}$$

is not equal to (a, b, c > 0).

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$

(C) $\pi$ (D) 0

**Q4.** If the numerical value of

$\tan\left\{\cos^{-1}\left(\frac{4}{5}\right)+\tan^{-1}\left(\frac{2}{3}\right)\right\}$ is $\frac{a}{b}$, then -

(A) a + b = 23 (B) a – b = 11

(C) 3b = a + 1 (D) 2a = 3b

**Q5.** If $\cos^{-1}x=\tan^{-1}x$, then

(A) $x^2=\left(\sqrt{5}-1\right)/2$

(B) $x^2=\left(\sqrt{5}+1\right)/2$

(C) $\sin\left(\cos^{-1}x\right)=\left(\sqrt{5}-1\right)/2$

(D) $\tan\left(\cos^{-1}x\right)=\left(\sqrt{5}-1\right)/2$

**Q6.** The solution of the equation $\sin\left[2\cos^{-1}\left\{\cot\left(2\tan^{-1}x\right)\right\}\right]=0$ are

(A) $\pm1$

(B) $1\pm\sqrt{2}$

(C) $-1\pm\sqrt{2}$

(D) $2\pm\sqrt{2}$

**Q7.** If $(\sin^{-1}x)^2+(\cos^{-1}x)^2=\frac{5\pi^2}{8}$, then x is equal to

(A) 1 (B) –1

(C) $\frac{1}{\sqrt{2}}$ (D) $-\frac{1}{\sqrt{2}}$

**Q8.** The values of x satisfying $\sin^{-1}x+\sin^{-1}(1-x)=\cos^{-1}x$ are

(A) 0 (B) $\frac{1}{2}$

(C) 1 (D) 2

**Q9.** $\sin^{-1}x>\cos^{-1}x$ holds for

(A) all values of x (B) $x\in\left(0,\frac{1}{\sqrt{2}}\right)$

(C) $x\in\left(\frac{1}{\sqrt{2}},1\right)$ (D) x = 0.75

## MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

**Q10.** **Column I** contains functions and **column II** contains their range. Match the entries of **column I** with the entries of **column II**.

| **Column-I** | **Column-II** |
|---|---|
| (A) $f(x)=\sin^{-1}\left(\frac{x}{1+\lvert x\rvert}\right)$ | (P) $(0,\pi)$ |
| (B) $g(x)=\cos^{-1}\left(\frac{x}{1+\lvert x\rvert}\right)$ | (Q) $\left(\frac{\pi}{4},\frac{3\pi}{4}\right)$ |
| (C) $h(x)=\tan^{-1}\left(\frac{x}{1+\lvert x\rvert}\right)$ | (R) $\left(-\frac{\pi}{4},\frac{\pi}{4}\right)$ |
| (D) $k(x)=\cot^{-1}\left(\frac{x}{1+\lvert x\rvert}\right)$ | (S) $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ |

**Q11.** **Column-I** **Column-II**

(A) $\sin(\tan^{-1}x)$ (P) $x$

(B) $\cos(\tan^{-1}x)$ (Q) $\frac{x}{\sqrt{x^2+1}}$

(C) $\sin(\cot^{-1}(\tan(\cos^{-1}x)))$.$x \in (0,1]$ (R) $\frac{1}{\sqrt{x^2+1}}$

(D) $\sin(\text{cosec}^{-1}(\cot(\tan^{-1}x)))$ ; $x \in (0,1]$ (S) $\sqrt{1-x^2}$

**Q12.** **Column-I** **Column-II**

(A) $\sin^{-1}\left(\sin\frac{33\pi}{7}\right)$ (P) $-2\pi/7$

(B) $\cos^{-1}\left(\cos\frac{46\pi}{7}\right)$ (Q) $2\pi/7$

(C) $\tan^{-1}\left(\tan\left(\frac{-33\pi}{7}\right)\right)$ (R) $3\pi/7$

(D) $\cot^{-1}\left(\cot\left(\frac{-46\pi}{7}\right)\right)$ (S) $4\pi/7$

## PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

**Paragraph # 1 (Ques. 13 to 15)**

Let $h_1(x) = \sin^{-1}(3x - 4x^3)$ ; $h_2(x) = \cos^{-1}(4x^3 - 3x)$ & $f(x) = h_1(x) + h_2(x)$

when $x \in [-1, \frac{-1}{2}]$ ;

let $f(x) = a\cos^{-1}x + b\pi$ ; $a, b \in Q$

$h_1(x) = p\sin^{-1}x + q\pi$ ; $p, q \in Q$

$h_2(x) = r\cos^{-1}x + s\pi$ ; $r, s \in Q$

Let $C_1$ be the circle with centre (p, q) & radius 1 & $C_2$ be the circle with centre (r, s) & radius 1.

**Q13.** $p + r + 2q - s =$

(A) 0 (B) 1

(C) 2 (D) 4

**Q14.** If $b.\log_{|s|}|p + q| = k.a$, then value of k is -

(A) $\frac{9}{2}$ (B) 6

(C) $\frac{-3}{2}$ (D) $-\frac{8}{3}$

**Q15.** Radical axis of circle $C_1$ & $C_2$ is -

(A) $12x - 2y - 3 = 0$

(B) $12x + 2y - 3 = 0$

(C) $-12x + 2y - 3 = 0$

(D) none of these

**Paragraph # 2 (Ques. 16 to 18)**

Consider the two equations in x ;

(i) $\sin\left(\frac{\cos^{-1}x}{y}\right)=1$ (ii) $\cos\left(\frac{\sin^{-1}x}{y}\right)=0$

The sets $X_1, X_2 \subseteq [-1, 1]$ ; $Y_1, Y_2 \subseteq I - \{0\}$ are such that

$X_1$ : the solution set of equation (i)

$X_2$ : the solution set of equation (ii)

$Y_1$ : the set of all integral values of y for which equation (i) possess a solution

$Y_2$ : the set of all intergral values of y for which equation (ii) possess a solution

Let : $C_1$ be the correspondence : $X_1 \to Y_1$ such that $x\ C_1 y$ for $x \in X_1$, $y \in Y_1$ & (x, y) satisfy (i).

$C_2$ be the correspondence : $X_2 \to Y_2$ such that $x\ C_2 y$ for $x \in X_2$, $y \in Y_2$ & (x, y) satisfy (ii).

**Q16.** The number of ordered pair (x, y) satisfying correspondence $C_1$ is

(A) 1 (B) 2

(C) 3 (D) 4

**Q17.** The number of ordered pair (x, y) satisfying correspondence $C_2$ is

(A) 1 (B) 2

(C) 3 (D) 4

**Q18.** $C_1 : X_1 \to Y_1$ is a function which is -

(A) one-one (B) many-one

(C) onto (D) into

## NUMERICAL TYPE

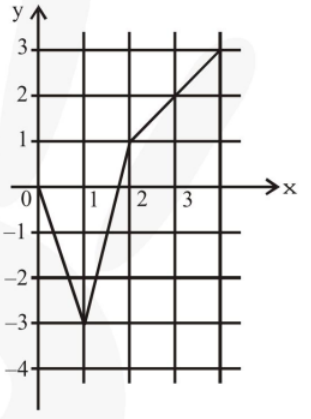
*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

**Q19.** Given is a partial graph of an even periodic function $f$ whose period is 8.

If [*] denotes greatest integer function then find the value of the expression.

$$f(-3) + 2|f(-1)| + \left[f\left(\frac{7}{8}\right)\right] + f(0)$$

$$+ \arccos(f(-2)) + f(-7) + f(20)$$




**Q20.** Let $y = \sin^{-1}(\sin 8) - \tan^{-1}(\tan 10) + \cos^{-1}(\cos 12) - \sec^{-1}(\sec 9) + \cot^{-1}(\cot 6) - \operatorname{cosec}^{-1}(\operatorname{cosec} 7)$. If y simplifies to $a\pi + b$, then find $(a - b)$.

**Q21.** Let $\sum_{k=1}^{\infty}\cot^{-1}\left(\frac{k^2}{8}\right) = \left(\frac{p}{q}\right)\pi$ where $\frac{p}{q}$ is rational in its lowest form. Find the value of $(p + q)$.

**Q22.** Let $\cos^{-1}x + \cos^{-1}(2x) + \cos^{-1}(3x) = \pi$. If x satisfies the cubic $ax^3 + bx^2 + cx - 1 = 0$, then find the value of $a + b + c$.

**Q23.** The value of $\tan\left(\sum_{k=1}^{10}\cot^{-1}(1 + k + k^2)\right) = \frac{a}{b}$ where a and b are coprime, find the value of $(a + b)$.

**Q24.** If the total area between the curves $f(x) = \cos^{-1}(\sin x)$ and $g(x) = \sin^{-1}(\cos x)$ on the interval $[-7\pi, 7\pi]$ is A, find the value of 49A. (Take $\pi = 22/7$).

## SUBJECTIVE TYPE

**Q25.** Find the domain and range of the given function.

(Read the symbols [ * ] and { * } as greatest integers and fractional part function respectively)

$$f(x) = 2^{\cos^{-1}\left(\sin\left(x+\frac{\pi}{3}\right)\right)} + \left[\frac{\sqrt{1-2\cos x}}{2}\right].$$

**Q26.** Find the domain of definition the following functions:

(Read the symbols [ * ] and { * } as greatest integers and fractional part functions respectively)

(a) $e^{\cos^{-1}x} + \cot^{-1}\left[\frac{x}{2}-1\right] + \frac{1}{2}\ \ln\{x\}$

(b) $f(x) = \frac{\sqrt{1-\sin x}}{\log_5(1-4x^2)} + \cos^{-1}(1-\{x\})$

**Q27.** Find all values of k for which there is a triangle whose angles have measure

$\tan^{-1}\left(\frac{1}{2}\right)$, $\tan^{-1}\left(\frac{1}{2}+k\right)$

and $\tan^{-1}\left(\frac{1}{2}+2k\right)$.

**Q28.** Solve the given equation :

$$2\tan^{-1}x = \cos^{-1}\frac{1-a^2}{1+a^2} - \cos^{-1}\frac{1-b^2}{1+b^2}$$

$a > 0,\ \ b > 0$

**Q29.** Find all the positive integral pair (x, y) satisfying the equation

$$\tan^{-1}x + \cos^{-1}\frac{y}{\sqrt{1+y^2}} = \sin^{-1}\frac{3}{\sqrt{10}}.$$

**Q30.** Solve the given inequality :

$4 \text{ arc } \tan^2 x - 8 \text{ arc } \tan x + 3 < 0$

& $4 \text{ arc } \cot x - \text{arc } \cot^2 x - 3 \geq 0$

**Q31.** Solve for x :

$$\sin^{-1}\left(\sin\left(\frac{2x^2+4}{1+x^2}\right)\right) < \pi - 3$$

**Q32.** Find the sum of the series :

(a) $\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{2}{9} + \ldots\ldots$

$+ \tan^{-1}\frac{2^{n-1}}{1+2^{2n-1}} + \ldots\ldots\infty$

(b) $\cot^{-1}7 + \cot^{-1}13 + \cot^{-1}21 + \cot^{-1}31 + \ldots.$

to n terms.

**Q33.** Express $f(x) = \arccos x + \arccos\left(\frac{x}{2} + \frac{1}{2}\sqrt{3-3x^2}\right)$ in simplest form and hence find the values of :

(a) $f\left(\frac{2}{3}\right)$ (b) $f\left(\frac{1}{3}\right)$

**Q34.** If $\sin^2 x + \sin^2 y < 1$ for all $x, y \in R$ then prove that $\sin^{-1}(\tan x \cdot \tan y) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

**Q35.** Find the set of values of 'a' for which the equation $2\cos^{-1}x = a + a^2(\cos^{-1}x)^{-1}$ posses a solution.

**Q36.** Express

$$\frac{\beta^3}{2}\operatorname{cosec}^2\left[\frac{1}{2}\tan^{-1}\frac{\beta}{\alpha}\right] + \frac{\alpha^3}{2}\sec^2\left[\frac{1}{2}\tan^{-1}\frac{\alpha}{\beta}\right]$$

as an integral polynomial in $\alpha$ & $\beta$.

# Answer Key

| Ex - 1 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| A | C | B | C | C | B | B | B | A | A |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| A | A | A | B | A | D | B | C | B | A |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| C | B | C | A | B | D | D | A | D | A |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| D | A | D | A | D | B | B | D | D | C |
| 41 | 42 | 43 | 44 | | | | | | |
| D | B | A | B | | | | | | |

## Ex - 1A

**1.** B,D **2.** A **3.** B **4.** C **5.** A,B,C,D

**6.** B,D **7.** B **8.** D **9.** C **10.** A,C

**11.** D **12.** C **13.** D **14.** D **15.** B

**16.** C

**17.** (A) $\rightarrow$ (Q); (B) $\rightarrow$ (R); (C) $\rightarrow$ (P)

**18.** (A) $\rightarrow$ (P); (B) $\rightarrow$ (Q,R); (C) $\rightarrow$ (P); (D) $\rightarrow$ (Q,S)

**19.** $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)=\frac{\pi}{6}$, $\cos^{-1}\left(-\frac{1}{2}\right)=\frac{2\pi}{3}$

**20.** $\cot^{-1}\left(\frac{-1}{\sqrt{3}}\right)=\frac{2\pi}{3}$

**22.** (i) 4/5 (ii) 4/5 (iii) $\frac{\sqrt{3}}{2}$ (iv) $-\frac{7}{17}$

| Ex - 2 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| B | D | A | C | A | C | B | C | D | B |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| B | D | B | B | C | A | C | B | B | D |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | |
| B | A | A | D | D | B | B | B | C | |

## Ex - 2A

**1.** B **2.** B **3.** A,B,D **4.** A, B, C **5.** A, C

**6.** A, B, C **7.** C, D **8.** A, B **9.** C, D

**10.** (A) → (S); (B) → (P); (C) → (R); (D) → (Q)

**11.** (A) → (Q), (B) → (R), (C) → (P), (D) → (P)

**12.** (A) → (Q), (B) → (S), (C) → (Q), (D) → (R)

**13.** B **14.** C **15.** A **16.** B **17.** D

**18.** A,C **19.** 5 **20.** 53 **21.** 11 **22.** 26

**23.** 11 **24.** 3388

**25.** $D=\left[\frac{\pi}{3},\frac{5\pi}{3}\right]=\left[2n\pi+\frac{\pi}{3},2n\pi+\frac{4\pi}{3}\right]$ $R=\left[2^{\pi/6},\ 2^{\pi}\right]$

**26.** (a) $(-1,\ 1)-\{0\}$ (b) $x \in (-1/2,\ 1/2),\ x \neq 0$

**27.** $k=\frac{11}{4}$ **28.** $x=\frac{a-b}{1+ab}$

**29.** $x = 1,\ y = 2$ and $x = 2,\ y = 7$

**30.** $\tan\frac{1}{2}<x\le\cot 1$ **31.** $x\in(-1,1)$

**32. (a)** $\frac{\pi}{4}$ **(b)** $\text{arc cot}\left[\frac{2n+5}{n}\right]$

**33. (a)** $\frac{\pi}{3}$ **(b)** $2\cos^{-1}\frac{1}{3}-\frac{\pi}{3}$

**34.** $\sin^{-1}(ab) \in \left(\frac{-\pi}{2},\frac{\pi}{2}\right)$ **35.** $a \in [-2\pi,\ \pi]-\{0\}$

**36.** $(\alpha^2+\beta^2)\ (\alpha^2+\beta^2)$

## -: Important Notes & Formulas :-